



University of Dhaka
Department of Computer Science and Engineering
Professional Masters in Information and Cyber Security (PMICS)
Semester: January, 2026

Adversary Emulation & Detection Lab Report

***Course Title:** Information Security and Cryptography*

***Course Code:** CSE 802, **Batch:** 6th*

Submitted To:

Mr. Mukul Ahmed

Teaching Assistant
Dept. of CSE, PMICS
University of Dhaka

Submitted By:

Md. Mainul Abedin Bhuyan

Dept. of CSE, PMICS
ID: 60036, Batch: 6th

Date of Submission: 10-06-2026

Github: <https://github.com/mainulabedin/Splunk-SIEM-Lab>

1. Introduction

This lab demonstrates the simulation and detection of various cyber attack techniques using Splunk Enterprise, Sysmon, Atomic Red Team and Splunk Universal Forwarder. The objective of this lab is to emulate real-world attacker behavior, collect security logs through Sysmon, forward the logs to Splunk Enterprise and create detection queries to identify malicious activities.

2. Objectives

The objectives of this lab were:

- Install and configure Splunk Enterprise.
- Configure Splunk Universal Forwarder.
- Install and configure Sysmon.
- Generate attack telemetry using Atomic Red Team.
- Collect and analyze logs in Splunk.
- Detect MITRE ATT&CK techniques using SPL queries.
- Identify Process Name, Command Line, User and Time information from logs.

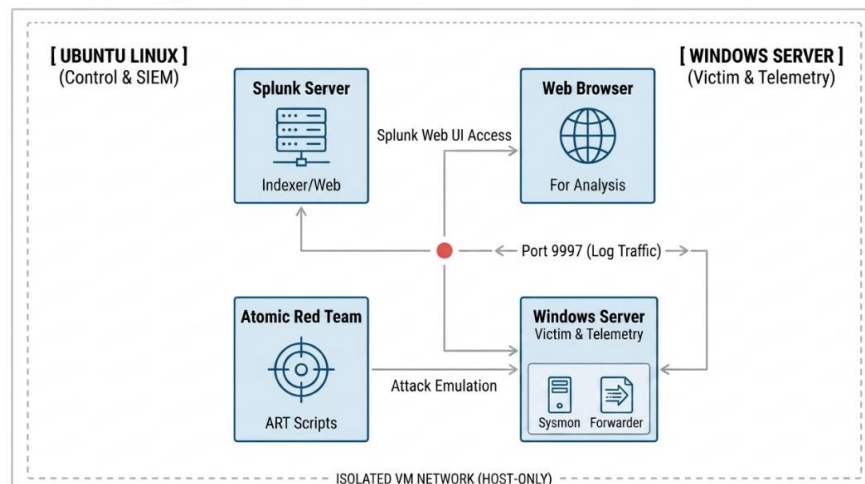
3. Lab Environment Setup

3.1 Lab Architecture

The lab consisted of two virtual machines:

Machine	Operating System	Purpose	CPU (Core)	RAM (GB)	Network Type
Splunk Server	Ubuntu Server (v20.04.6)	(SIEM)	2	4	NAT
Target Machine	Windows Server 2019	Attack Simulation	2	4	NAT

Lab Network Diagram:



3.2 Ubuntu Server Configuration:

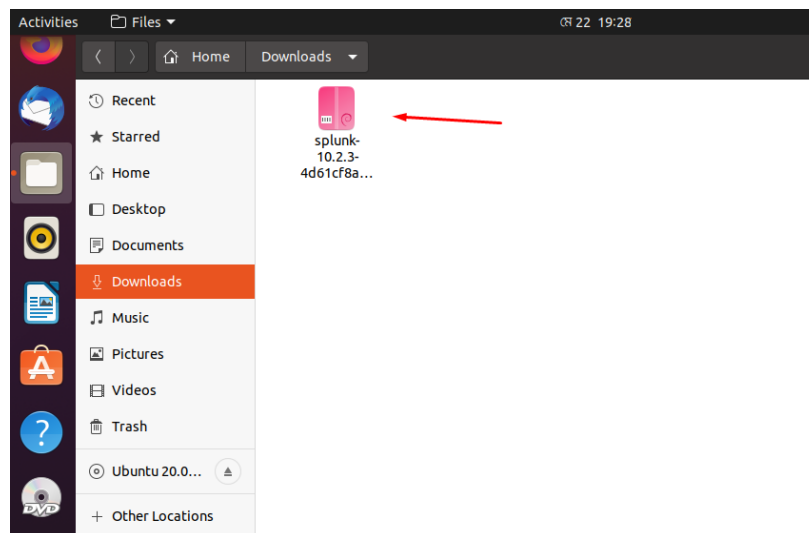
1. Download Splunk Enterprise from https://www.splunk.com/en_us/download/splunk-enterprise.html

Splunk Enterprise 10.4.0

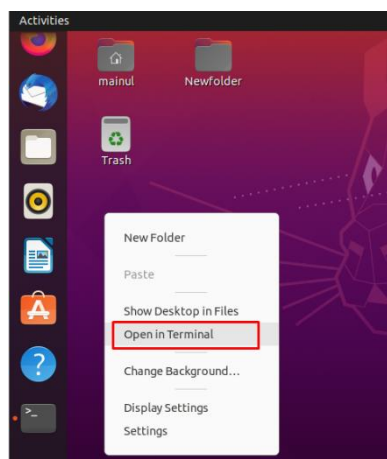
Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

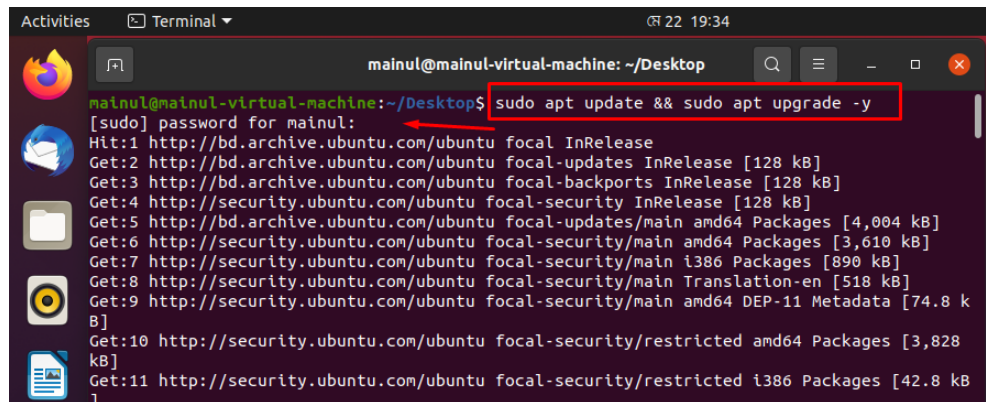
Choose Your Installation Package

 Windows	 Linux	 Mac OS				
64-bit	4.x+, 5.x+, 6.x+ kernel Linux distributions	.rpm	1506.12 MB	Download Now 	Copy wget link 	More 
		.deb	1240.27 MB	Download Now 	Copy wget link 	More 
		.tgz	1579.9 MB	Download Now 	Copy wget link 	More 



2. Now open a Terminal and run `sudo apt update && sudo apt upgrade -y` and enter your root password



A terminal window titled 'mainul@mainul-virtual-machine: ~/Desktop' with a search bar and window controls. The command 'sudo apt update && sudo apt upgrade -y' is entered and highlighted with a red box. The output shows the system updating various packages from the Ubuntu focal repository, including security updates and package metadata. The terminal background is dark purple with light blue and white text.

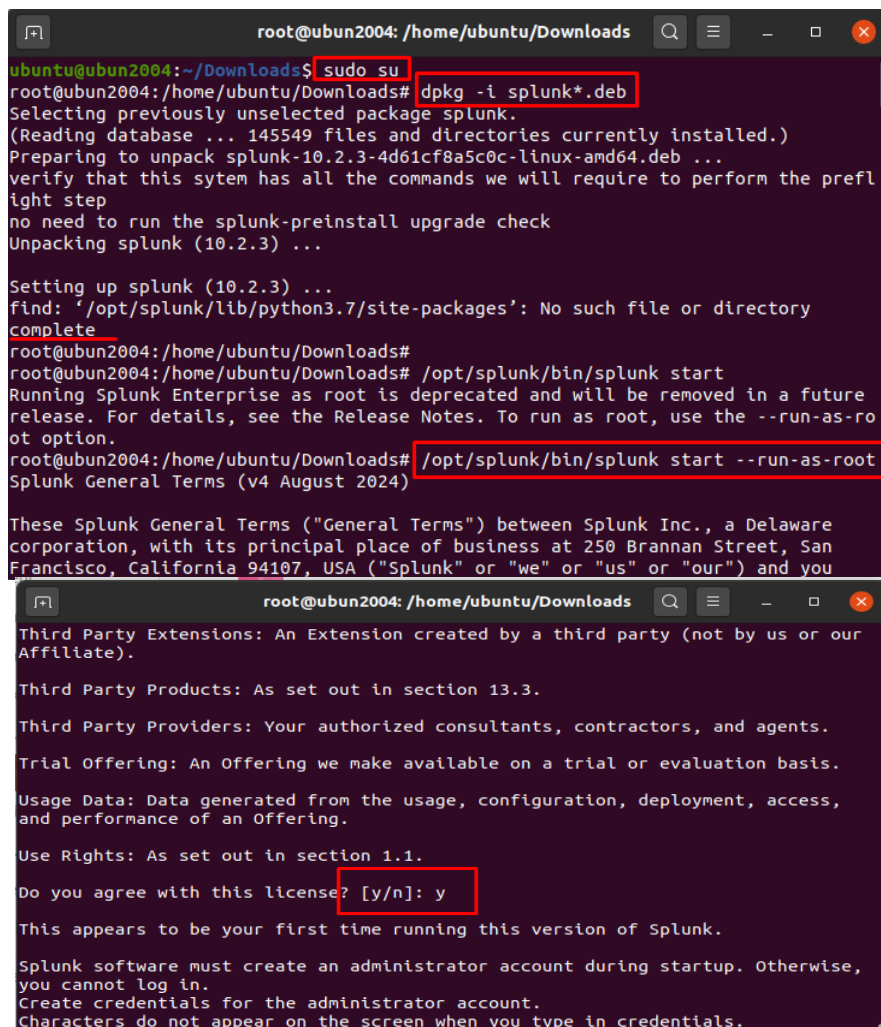
```
mainul@mainul-virtual-machine: ~/Desktop$ sudo apt update && sudo apt upgrade -y
[sudo] password for mainul:
Hit:1 http://bd.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://bd.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:3 http://bd.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:5 http://bd.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [4,004 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,610 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [890 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [518 kB]
Get:9 http://security.ubuntu.com/ubuntu focal-security/main amd64 DEP-11 Metadata [74.8 kB]
Get:10 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3,828 kB]
Get:11 http://security.ubuntu.com/ubuntu focal-security/restricted i386 Packages [42.8 kB]
```

3. Now open a **new Terminal in Downloads directory** and run →

```
sudo su
```

```
dpkg -i splunk*.deb
```

```
/opt/splunk/bin/splunk start --run-as-root
```

A terminal window titled 'root@ubun2004: /home/ubuntu/Downloads' with a search bar and window controls. The command 'sudo su' is entered and highlighted with a red box. The prompt changes to 'root@ubun2004: /home/ubuntu/Downloads#'. The command 'dpkg -i splunk*.deb' is entered and highlighted with a red box. The output shows the installation of Splunk 10.2.3, including unpacking and setting up. The command '/opt/splunk/bin/splunk start' is entered, and the output shows the start of Splunk Enterprise. The command '/opt/splunk/bin/splunk start --run-as-root' is entered and highlighted with a red box. The output shows the Splunk General Terms and the user's agreement to the license. The terminal background is dark purple with light blue and white text.

```
root@ubun2004: /home/ubuntu/Downloads$ sudo su
root@ubun2004: /home/ubuntu/Downloads# dpkg -i splunk*.deb
Selecting previously unselected package splunk.
(Reading database ... 145549 files and directories currently installed.)
Preparing to unpack splunk-10.2.3-4d61cf8a5c0c-linux-amd64.deb ...
verify that this system has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunk (10.2.3) ...

Setting up splunk (10.2.3) ...
find: '/opt/splunk/lib/python3.7/site-packages': No such file or directory
complete
root@ubun2004: /home/ubuntu/Downloads#
root@ubun2004: /home/ubuntu/Downloads# /opt/splunk/bin/splunk start
Running Splunk Enterprise as root is deprecated and will be removed in a future release. For details, see the Release Notes. To run as root, use the --run-as-root option.
root@ubun2004: /home/ubuntu/Downloads# /opt/splunk/bin/splunk start --run-as-root
Splunk General Terms (v4 August 2024)

These Splunk General Terms ("General Terms") between Splunk Inc., a Delaware corporation, with its principal place of business at 250 Brannan Street, San Francisco, California 94107, USA ("Splunk" or "we" or "us" or "our") and you

Third Party Extensions: An Extension created by a third party (not by us or our Affiliate).

Third Party Products: As set out in section 13.3.

Third Party Providers: Your authorized consultants, contractors, and agents.

Trial Offering: An Offering we make available on a trial or evaluation basis.

Usage Data: Data generated from the usage, configuration, deployment, access, and performance of an Offering.

Use Rights: As set out in section 1.1.

Do you agree with this license? [y/n]: y

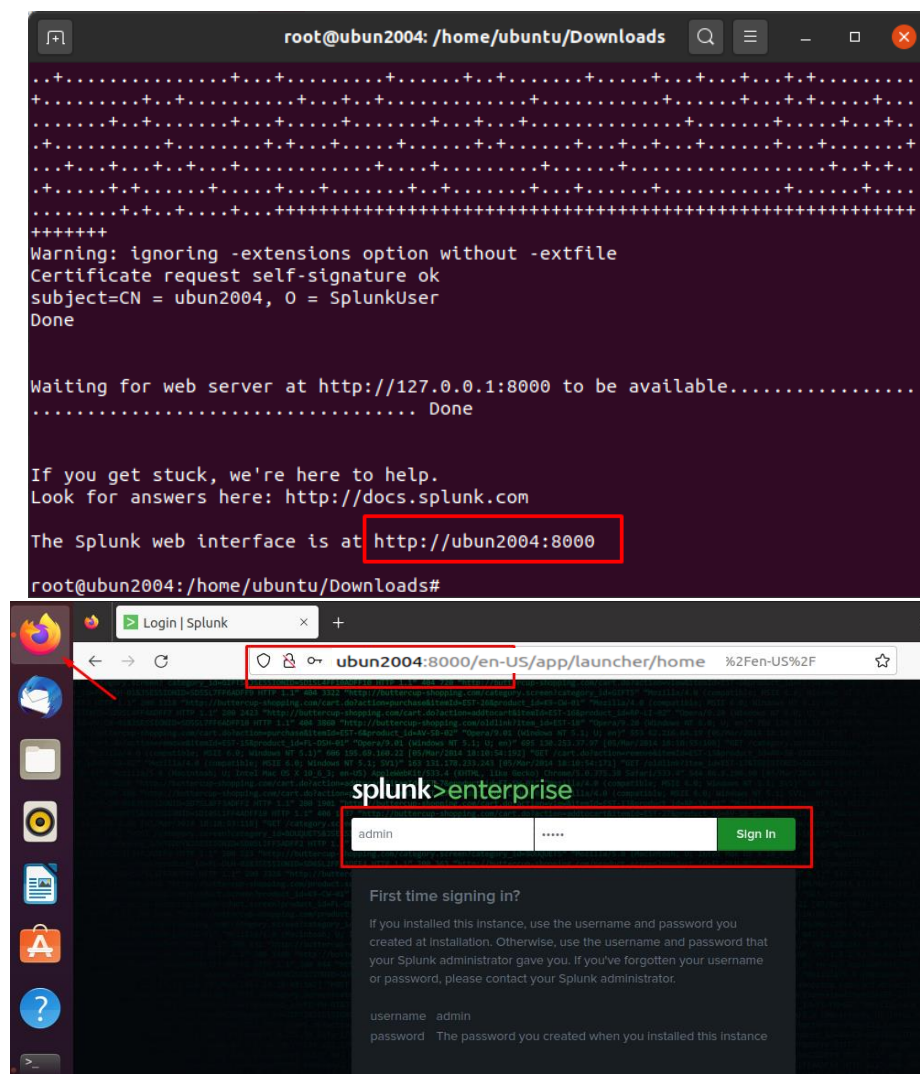
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.
```

4. Write a username and password for login Splunk server

```
root@ubun2004: /home/ubuntu/Downloads
Please enter an administrator username: admin
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
writing RSA key
writing RSA key
Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.
Splunk> Map. Reduce. Recycle.
```

5. Now copy this link (<http://ubun2004:8000>) and go to a **latest version browser**. Paste this link and enter the created username and password.



6. Now click **Settings** → **Forwarding and receiving** and configure receiving (port: 9997).

The screenshot shows the Splunk Enterprise web interface. The top navigation bar includes 'splunk>enterprise', 'Apps', and user options like 'Administrator', 'Messages', 'Settings' (highlighted with a red box), 'Activity', and 'Help'. The left sidebar shows 'Apps' with a search bar and categories like 'Search & Reporting', 'Audit Trail', 'Data Management', and 'Discover Splunk Observability Cloud'. The main content area is titled 'Hello, Administrator' and contains a 'Bookmarks' section. Below this, there are 'Add Data' and 'Explore Data' buttons. The 'Settings' section is divided into 'KNOWLEDGE' and 'DATA' categories. Under 'DATA', 'Forwarding and receiving' is highlighted with a red box. Other options include 'Data inputs', 'Indexes', 'Report acceleration summaries', 'Virtual indexes', 'Source types', and 'Ingest actions'.

This screenshot shows the 'Configure receiving' form. The title is 'Receive data' with a subtitle 'Configure this instance to receive data forwarded from other instances.' Below this is a table with columns 'Type' and 'Actions'. The 'Type' column contains 'Configure receiving', and the 'Actions' column has a '+ Add new' button (highlighted with a red box). The 'Configure receiving' section has a subtitle 'Set up this Splunk instance to receive data from forwarder(s).' and a form field 'Listen on this port' with the value '9997' (highlighted with a red box and an arrow). Below the field is a note: 'For example, 9997 will receive data on TCP port 9997.' At the bottom right are 'Cancel' and 'Save' buttons, with the 'Save' button highlighted by a red arrow.

This screenshot shows the 'Receive data' table after saving the configuration. The title is 'Receive data' with a subtitle 'Forwarding and receiving > Receive data'. A green button 'New Receiving Port' is in the top right. A blue banner at the top says 'Successfully saved "9997"'. Below this, it says 'Showing 1-1 of 1 item'. There is a search bar and a '25 per page' dropdown. The table has columns 'Listen on this port', 'Status', and 'Actions'. The first row shows '9997' in the 'Listen on this port' column, 'Enabled | Disable' in the 'Status' column, and 'Delete' in the 'Actions' column.

7. Create an **index** in splunk (**wineventlog**)

Run → `/opt/splunk/bin/splunk add index wineventlog`

Check created index → `/opt/splunk/bin/splunk list index`

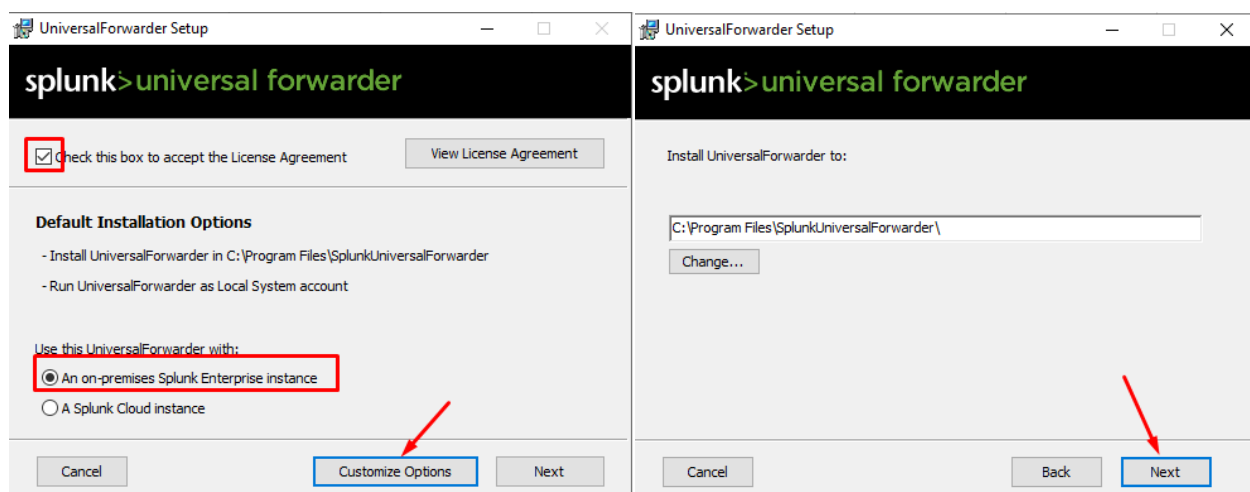
Now restart the splunk server → `/opt/splunk/bin/splunk restart --run-as-root`

```
root@ubun2004: /home/ubuntu/Downloads
root@ubun2004:/home/ubuntu/Downloads# /opt/splunk/bin/splunk add index wineventlog
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslCo
nfig]/cliVerifyServerName for details.
An index with name=wineventlog already exists
root@ubun2004:/home/ubuntu/Downloads# /opt/splunk/bin/splunk list index
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslCo
nfig]/cliVerifyServerName for details.
_audit
  /opt/splunk/var/lib/splunk/audit/db
  /opt/splunk/var/lib/splunk/audit/coldb
  /opt/splunk/var/lib/splunk/audit/thawedb
summary
  /opt/splunk/var/lib/splunk/summarydb/db
  /opt/splunk/var/lib/splunk/summarydb/coldb
  /opt/splunk/var/lib/splunk/summarydb/thawedb
wineventlog
  /opt/splunk/var/lib/splunk/wineventlog/db
  /opt/splunk/var/lib/splunk/wineventlog/coldb
  /opt/splunk/var/lib/splunk/wineventlog/thawedb
root@ubun2004:/home/ubuntu/Downloads# /opt/splunk/bin/splunk restart --run-as-root
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
.....
Stopping splunk helpers...
```

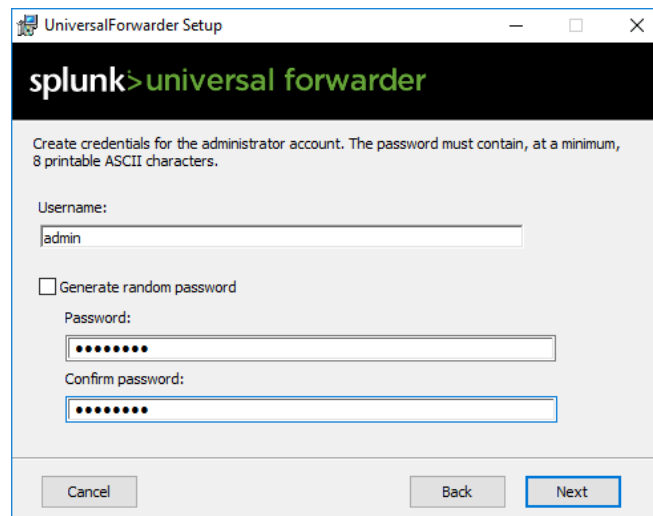
3.3 Windows Server Configuration

1. Download and Install Splunk forwarder form:

https://www.splunk.com/en_us/download/universal-forwarder.html



2. Enter a username and password for forwarder



UniversalForwarder Setup

splunk>universal forwarder

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

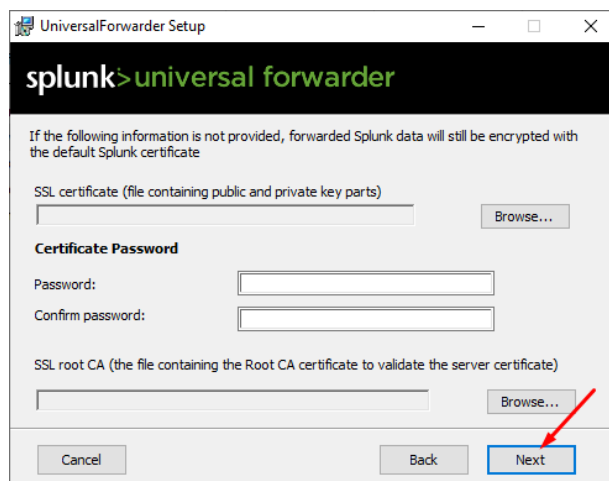
Username:
admin

☐ Generate random password

Password:
••••••••

Confirm password:
••••••••

Cancel Back Next



UniversalForwarder Setup

splunk>universal forwarder

If the following information is not provided, forwarded Splunk data will still be encrypted with the default Splunk certificate

SSL certificate (file containing public and private key parts)
Browse...

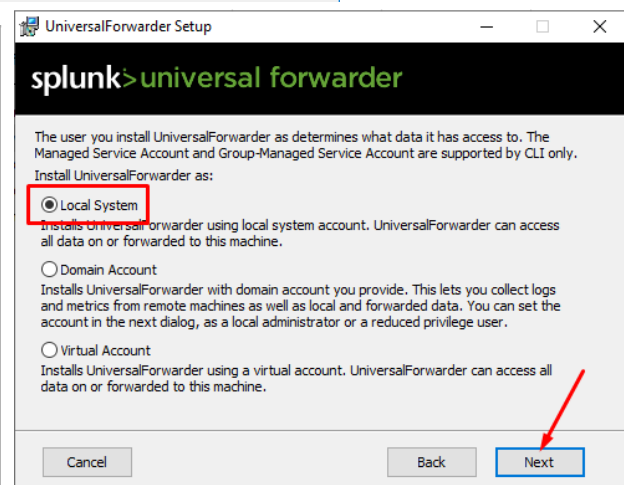
Certificate Password

Password:

Confirm password:

SSL root CA (the file containing the Root CA certificate to validate the server certificate)
Browse...

Cancel Back Next



UniversalForwarder Setup

splunk>universal forwarder

The user you install UniversalForwarder as determines what data it has access to. The Managed Service Account and Group-Managed Service Account are supported by CLI only.

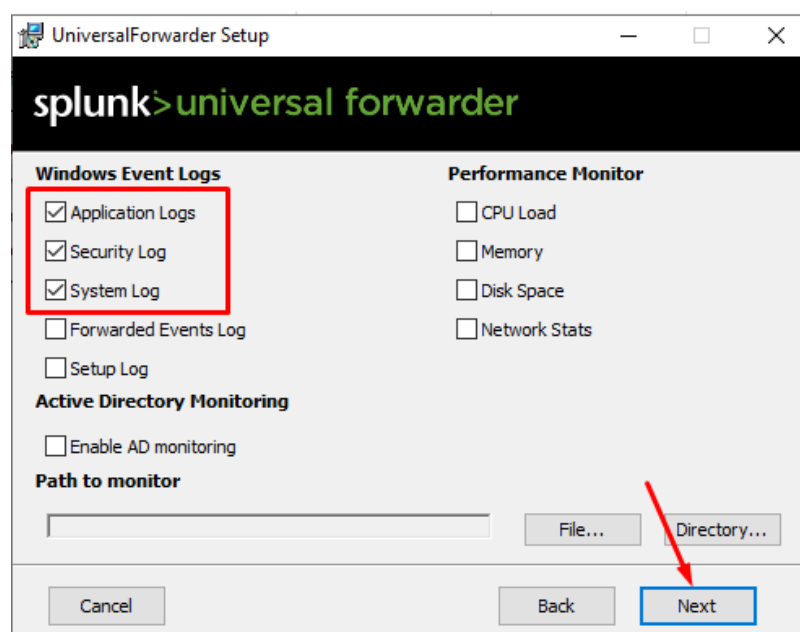
Install UniversalForwarder as:

☒ Local System
Installs UniversalForwarder using local system account. UniversalForwarder can access all data on or forwarded to this machine.

☐ Domain Account
Installs UniversalForwarder with domain account you provide. This lets you collect logs and metrics from remote machines as well as local and forwarded data. You can set the account in the next dialog, as a local administrator or a reduced privilege user.

☐ Virtual Account
Installs UniversalForwarder using a virtual account. UniversalForwarder can access all data on or forwarded to this machine.

Cancel Back Next



UniversalForwarder Setup

splunk>universal forwarder

Windows Event Logs

☒ Application Logs

☒ Security Log

☒ System Log

☐ Forwarded Events Log

☐ Setup Log

Active Directory Monitoring

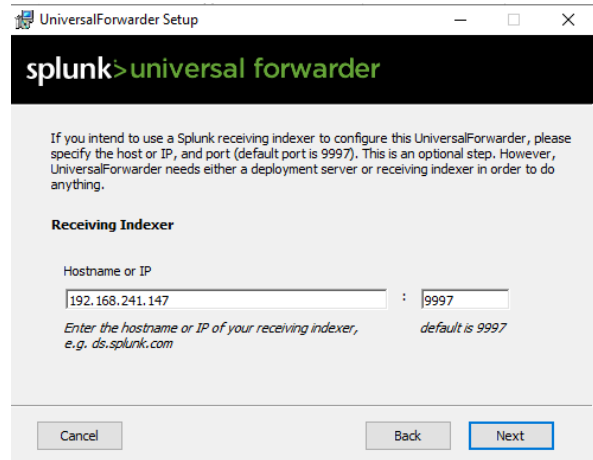
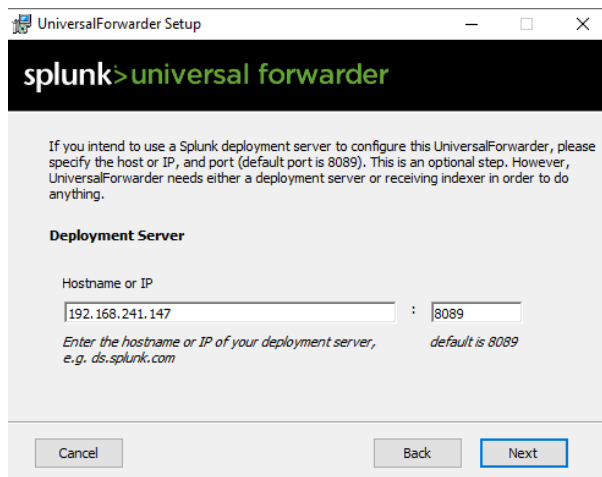
☐ Enable AD monitoring

Path to monitor

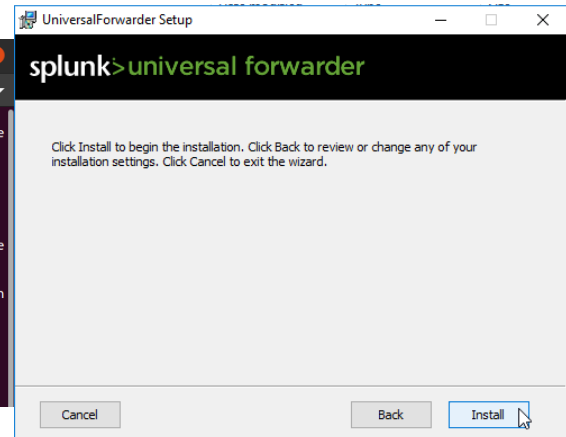
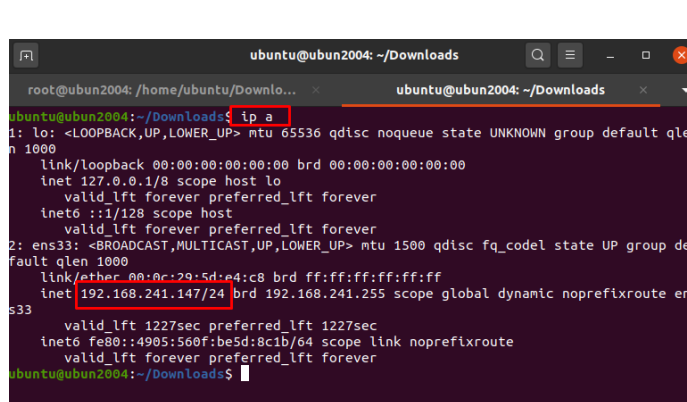
File... Directory...

Cancel Back Next

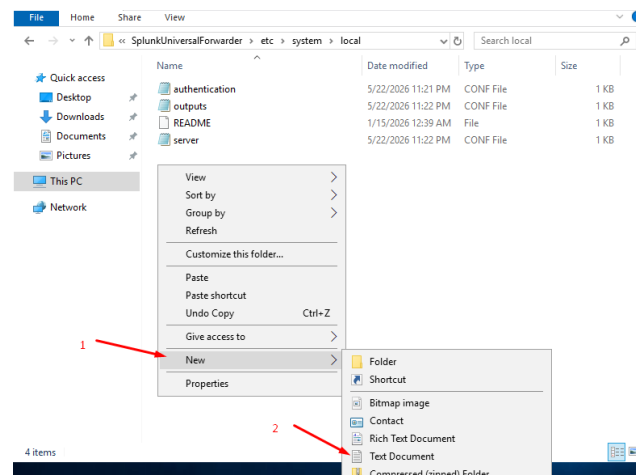
3. Enter Splunk server IP (Ubuntu server IP)

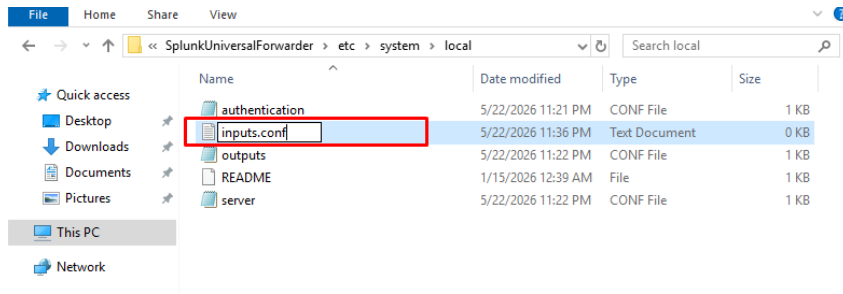


Check Ubuntu server IP:

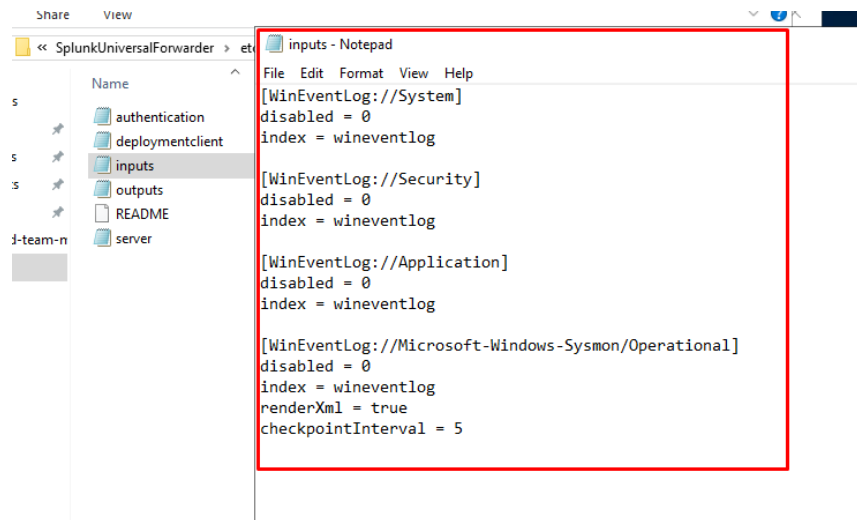


4. Now go to Local Disk (C:) > Program Files > SplunkUniversalForwarder > etc > system > local > and Create an inputs.conf file

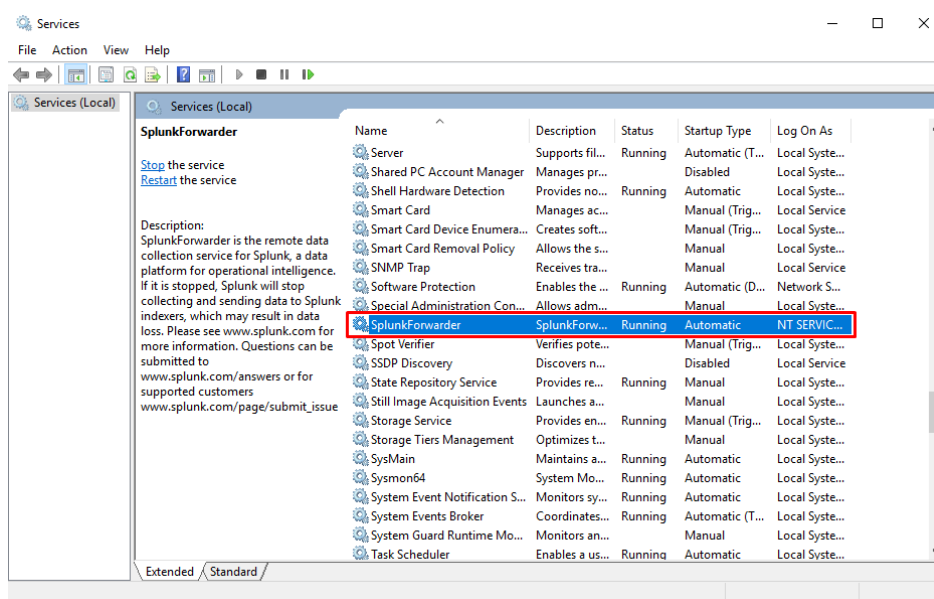




5. Open **inputs.conf** file and edit as image instruction and save →



6. Check “SplunkForwarder status running” in **services** →



7. Now download **Sysmon** and setup from: <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>

Learn > Sysinternals > Ask Learn

Sysmon v15.2

Summarize this article for me

By Mark Russinovich and Thomas Garnier

Published: March 26, 2026

 Download Sysmon [↗] (4.6 MB)

[Download Sysmon for Linux \(GitHub\) [↗]](#)

8. Download **sysmonconfig.xml** from: <https://github.com/swiftonsecurity/sysmon-config>

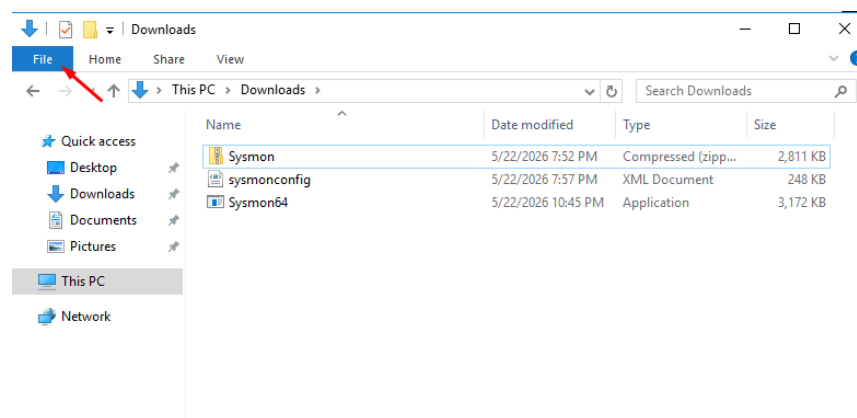
SwiftOnSecurity / sysmon-config Public

[Code](#) [Issues 54](#) [Pull requests 28](#) [Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#)

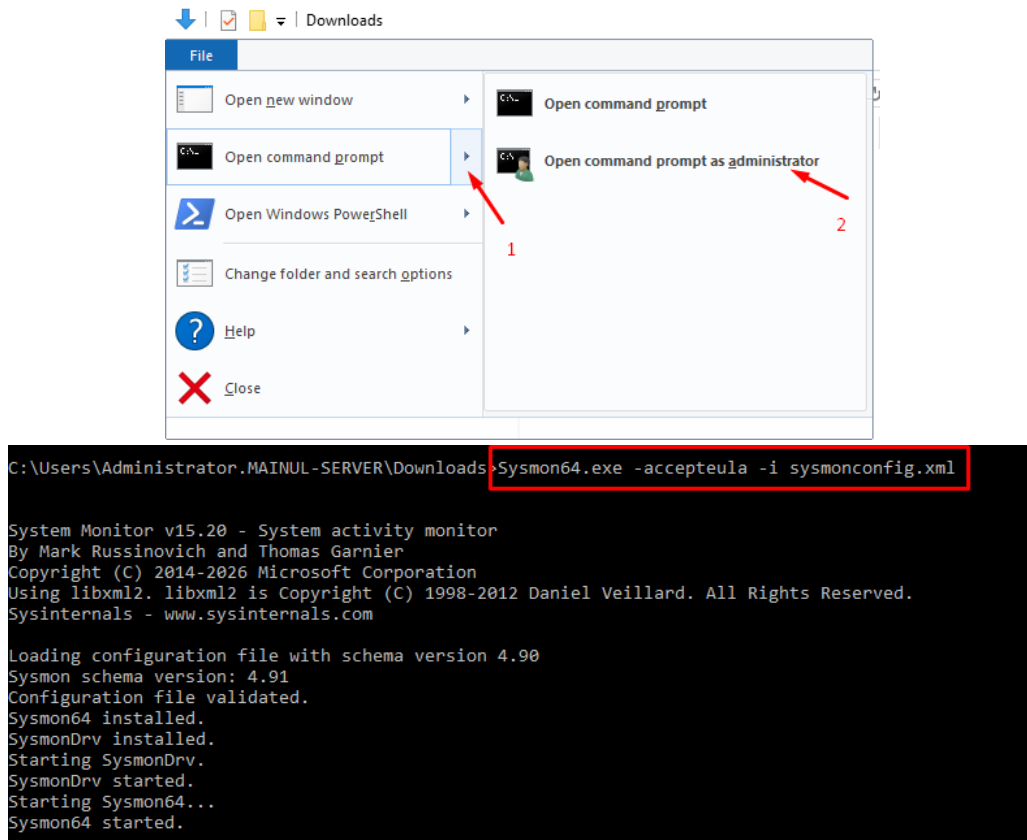
master 1 Branch 0 Tags Code

SwiftOnSecurity Merge pull request #151 from Neo23x0/patch-8		1836897 · 5 years ago	173 Commits
	.gitignore	d	7 years ago
	README.md	Update README.md	5 years ago
	sysmonconfig-export.xml	Merge pull request #151 from Neo23x0/patch-8	5 years ago

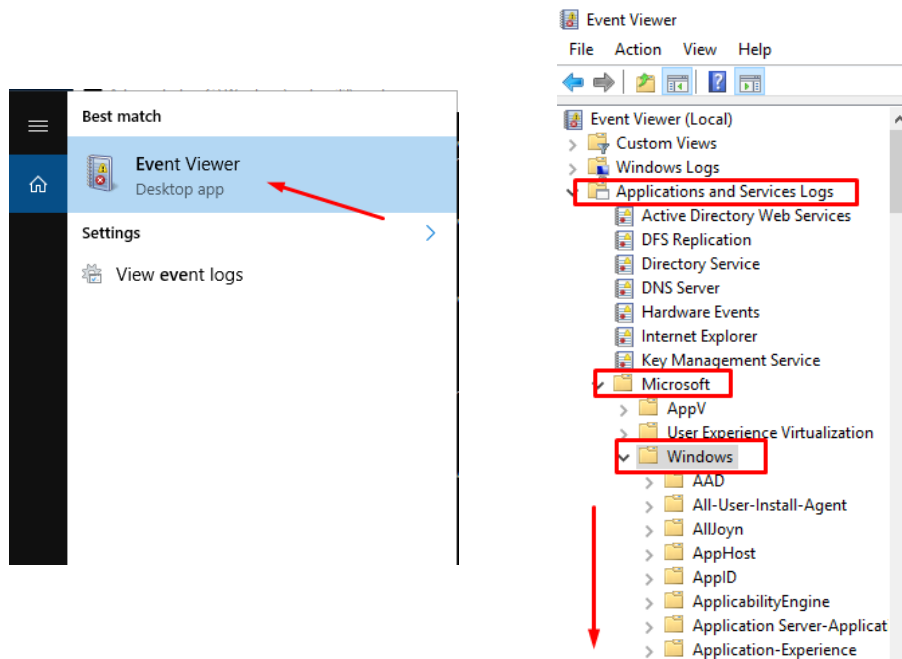
9. Now go to downloads folder, then extract **Sysmon Zip** file, then copy **Sysmon64.exe** and paste downloads folder and follow the steps:



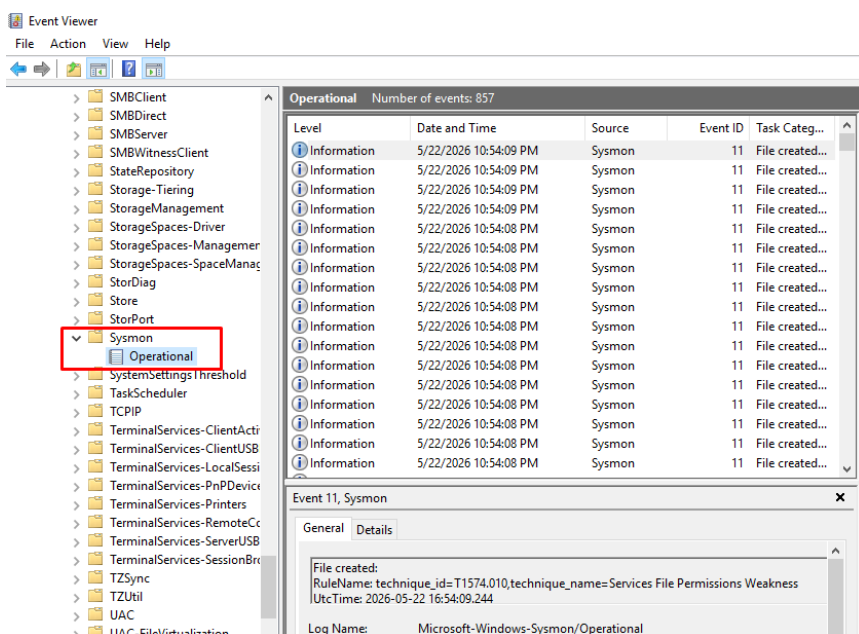
10. Now open cmd run as administrator and install **Sysmon64** →



11. Now check Sysmon successful installation. Go to Event Viewer →



Sysmon installed successfully and store event

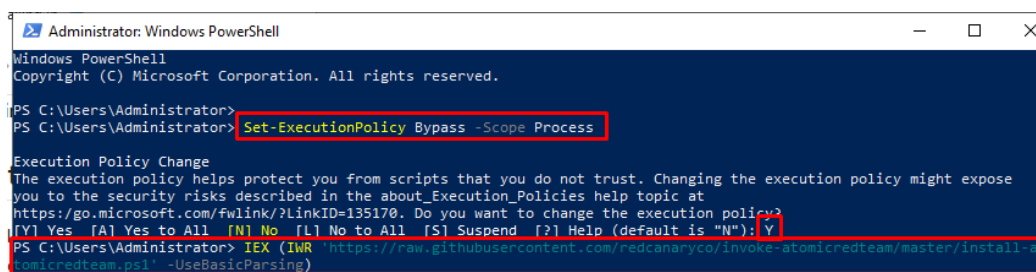


3.4 Atomic Red Team Setup

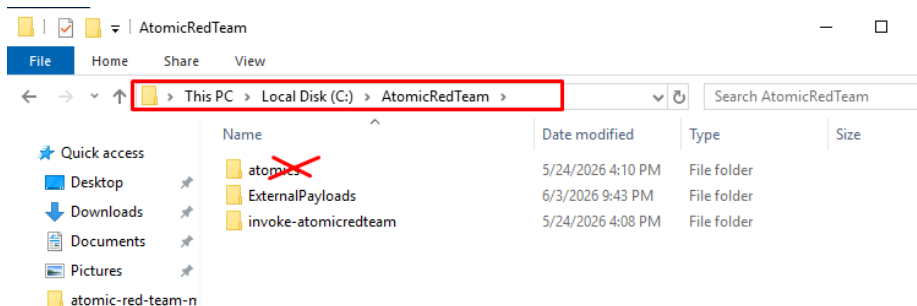
1. Now open Powershell run as administrator and run the following command →

Set-ExecutionPolicy Bypass -Scope Process

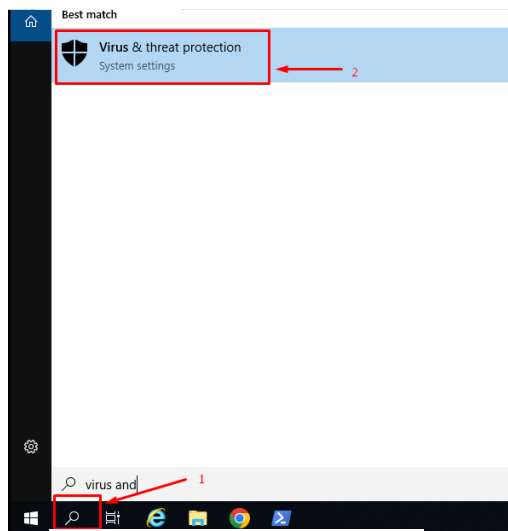
IEX (IWR 'https://raw.githubusercontent.com/redcanaryco/atomicredteam/master/install-atomicredteam.ps1' -UseBasicParsing)



2. Go to “Local Disk (C:) > AtomicRedTeam” folder. If there not found “atomics” folder following the next instruction (3 to 5), otherwise skip.



3. Open virus & threat protection and turn off protection →



Real-time protection

Locates and stops malware from installing or running on your device. You can turn off this setting for a short time before it turns back on automatically.

Real-time protection is off, leaving your device vulnerable.

Off

Cloud-delivered protection

Provides increased and faster protection with access to the latest protection data in the cloud. Works best with Automatic sample submission turned on.

Cloud-delivered protection is off. Your device may be vulnerable.

Off

[Privacy Statement](#)



Automatic sample submission

Send sample files to Microsoft to help protect you and others from potential threats. We'll prompt you if the file we need is likely to contain personal information.

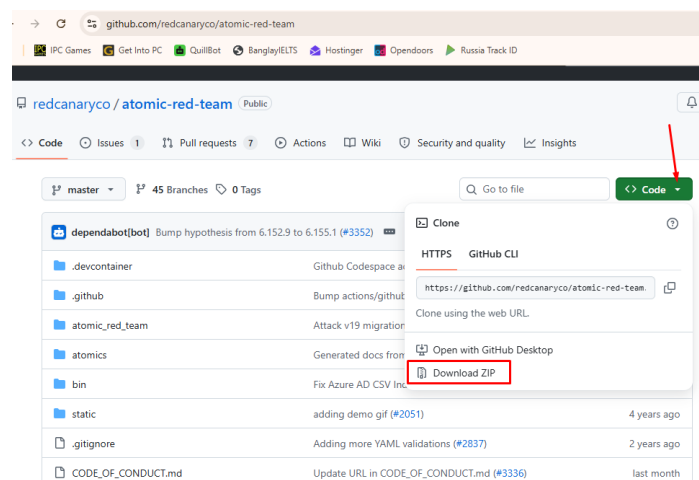
Automatic sample submission is off. Your device may be vulnerable.

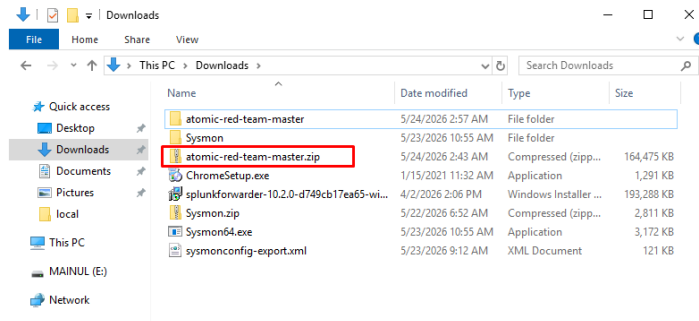
Off

[Privacy Statement](#)

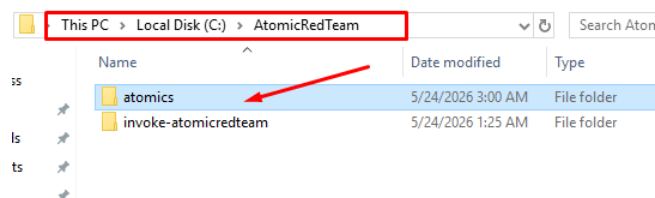
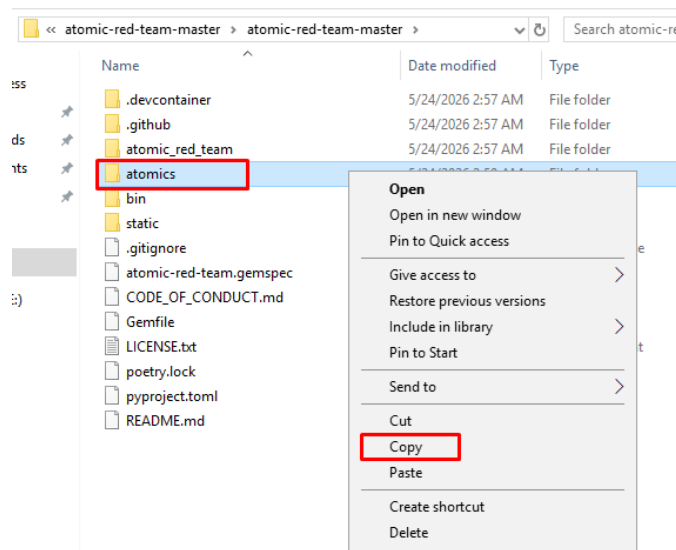
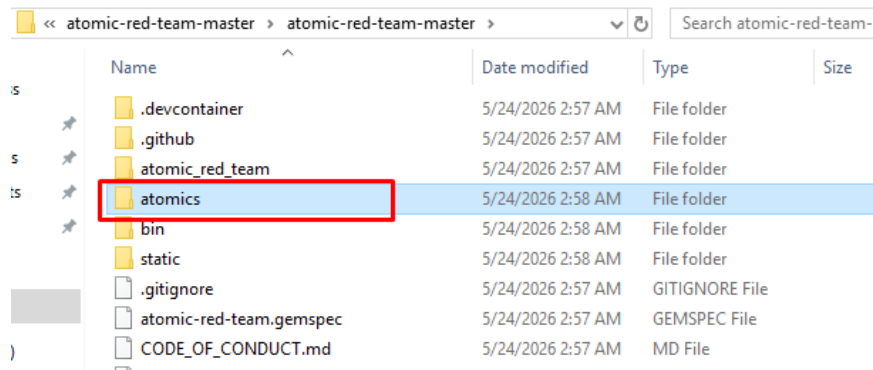
[Submit a sample manually](#)

4. Now download atomic-red-team zip file from: <https://github.com/redcanaryco/atomic-red-team> and extract the zip file.





5. Go to **atomic-red-team-master** folder and copy “**atomics**” folder and paste it in “**Local Disk (C:) > AtomicRedTeam**” folder →



6. Now install atomic red team →

Install-AtomicRedTeam

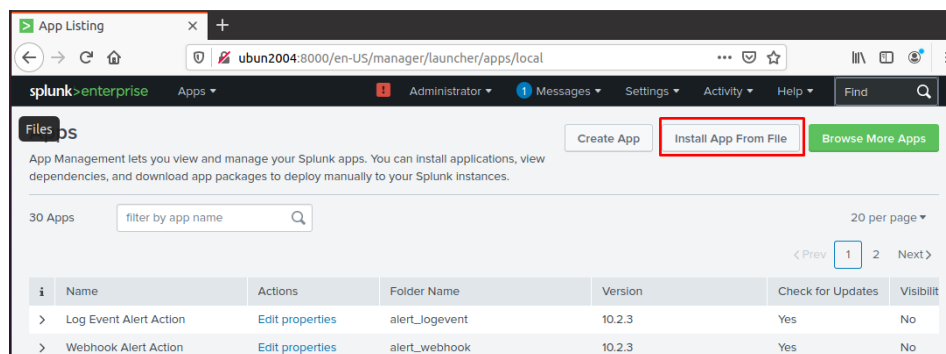
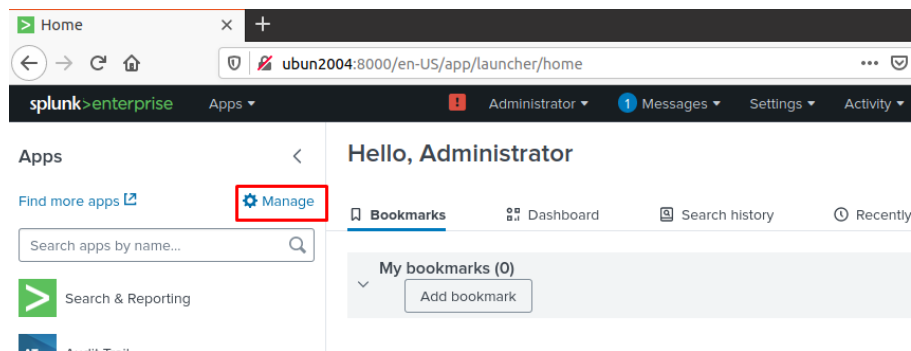
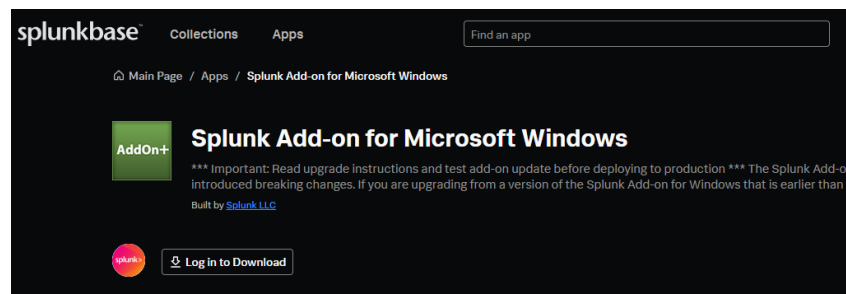
Install-Module -Name Invoke-AtomicRedTeam -Force

Import-Module Invoke-AtomicRedTeam

```
PS C:\Users\Administrator> Install-AtomicRedTeam
Installation of Invoke-AtomicRedTeam is complete. You can now use the Invoke-AtomicTest function
See Wiki at https://github.com/redcanaryco/atomicredteam/wiki for complete details

PS C:\Users\Administrator> Install-Module -Name Invoke-AtomicRedTeam -Force
PS C:\Users\Administrator> Import-Module Invoke-AtomicRedTeam
```

7. **Ubuntu Server Machine:** download **Splunk Add-on for Microsoft Windows** from: <https://splunkbase.splunk.com/> and install it following instruction →



Install App From File

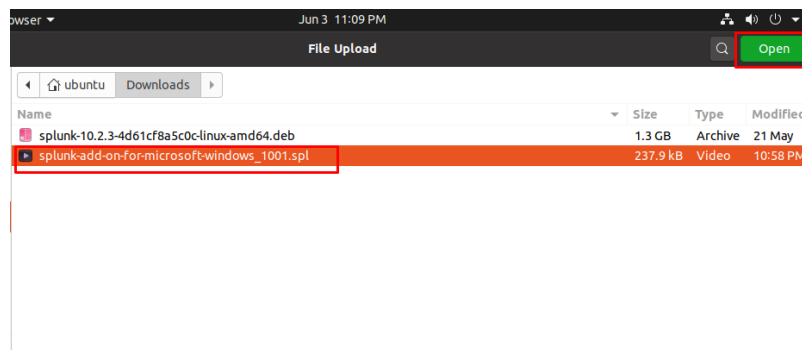
Supported app package file formats are .spl, .tar, .tar.gz or .tgz file.

Drop your file here or browse...

☐ Upgrade app. Checking this will overwrite the app if it already exists.

Close

Upload



Install App From File

Supported app package file formats are .spl, .tar, .tar.gz or .tgz file.

Drop your file here or browse...

splunk-add-on-for-microsoft-windows_1001.spl

☐ Upgrade app. Checking this will overwrite the app if it already exists.

Close

Upload

Install - Success

Splunk Add-on for Microsoft Windows has been successfully installed.

Close

4. Attack Simulation and Detection

Attack 1: T1053.005 (Persistence) | Scheduled Task

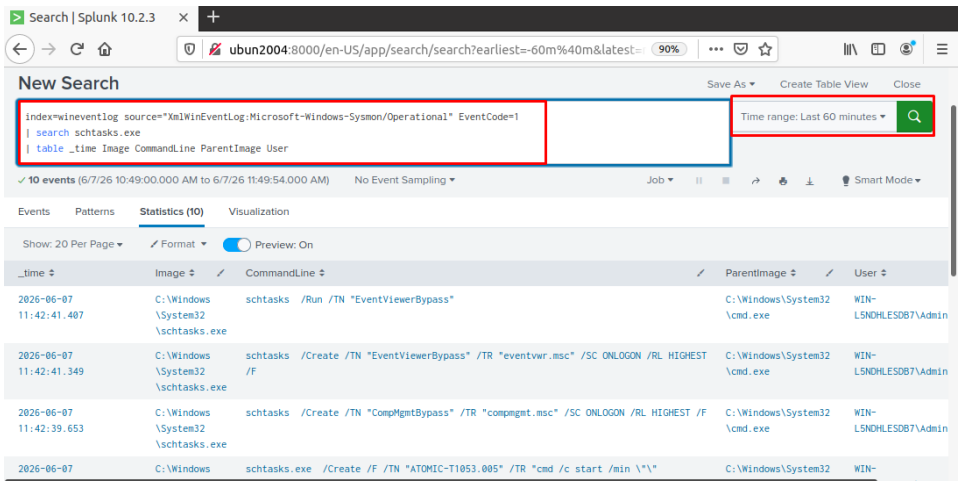
A malicious scheduled task was created using Atomic Red Team.

```
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005
PathToAtomicFolder = C:\AtomicRedTeam\atomicms

Executing test: T1053.005-1 Scheduled Task Startup Script
SUCCESS: The scheduled task "T1053_005_OnLogon" has successfully been created.
SUCCESS: The scheduled task "T1053_005_OnStartup" has successfully been created.
Exit code: 0
Done executing test: T1053.005-1 Scheduled Task Startup Script
Executing test: T1053.005-2 Scheduled task Local
SUCCESS: The scheduled task "spawn" has successfully been created.
Exit code: 0
Done executing test: T1053.005-2 Scheduled task Local
Executing test: T1053.005-3 Scheduled task Remote
ERROR: No mapping between account names and security IDs was done.
Exit code: 1
Done executing test: T1053.005-3 Scheduled task Remote
Executing test: T1053.005-4 Powershell Cmdlet Scheduled Task
TaskPath TaskName State
-----
\ AtomicTask Ready
Exit code: 0
Done executing test: T1053.005-4 Powershell Cmdlet Scheduled Task
Executing test: T1053.005-5 Task Scheduler via VBA
New-Object : Retrieving the COM class factory for component with CLSID {00000000-0000-0000-0000-000000000000} failed
due to the following error: 80040154 Class not registered (Exception from HRESULT: 0x80040154 (REGDB_E_CLASSNOTREG)).
At line:70 char:12
```

Detection and Query:

index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
| search schtasks.exe
| table _time Image CommandLine ParentImage User

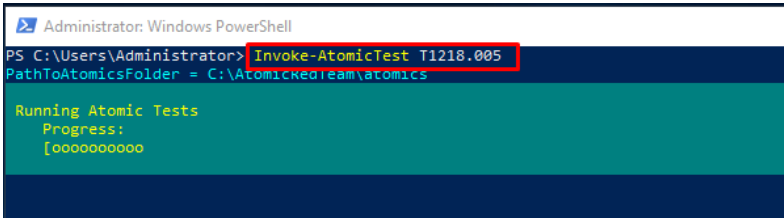


Sysmon Event ID: 1 (Process Creation)

Event ID 1 records process creation activities. It was the most useful event because it captured the execution of **schtasks.exe** along with the full command line, allowing identification of the malicious scheduled task creation.

Attack 2: **T1218.005 (Defense Evasion) | MSHTA**

A malicious HTA file was executed using mshta.exe.



Detection and Query:

index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
| search mshta.exe
| table _time Image CommandLine ParentImage User

A screenshot of the Windows Event Viewer interface. The search bar contains the query: 'index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1'. Below the search bar, there are tabs for 'Events', 'Patterns', 'Statistics (13)', and 'Visualization'. The 'Statistics (13)' tab is selected, showing a table of events. The table has columns: '_time', 'Image', 'CommandLine', 'ParentImage', and 'User'. There are three rows of data, all showing events related to mshta.exe execution. The first row shows a command line with 'Write-Host Hello, MSHTA!;Start-Sleep -Seconds 5'. The second row shows a command line with a script block that runs 'powershell.exe' and 'cmd.exe'. The third row shows a command line with a script block that runs 'cmd.exe' and 'powershell.exe'.

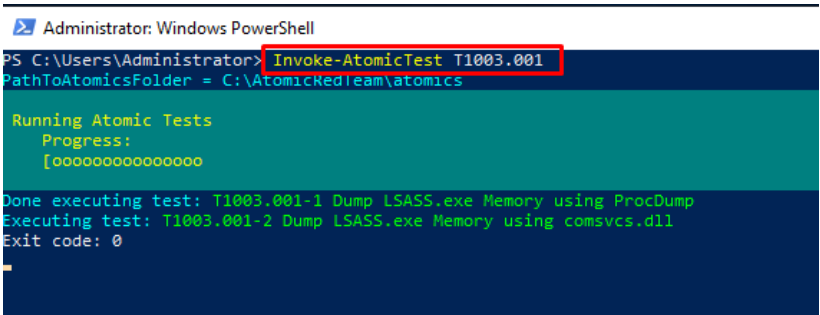
_time	Image	CommandLine	ParentImage	User
2026-06-07 11:57:02.056	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -nop -Command Write-Host Hello, MSHTA!;Start-Sleep -Seconds 5	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	WIN-L5NDHLESDB7\Adminis
2026-06-07 11:57:01.841	C:\Windows\System32\cmd.exe	mshta.exe "about:<hta:application<<script language="VBScript"<Close(Execute("CreateObject("Wscript.Shell").Run%%20"powershell.exe%%20-nop%%20-Command%%20Write-Host%%20Hello,%%20MSHTA!;Start-Sleep%%20-Seconds%%205"))</script<"	C:\Windows\System32\cmd.exe	WIN-L5NDHLESDB7\Adminis
2026-06-07 11:57:01.786	C:\Windows\System32\cmd.exe	"cmd.exe" /c mshta.exe "about:<hta:application<<script language="VBScript"<Close(Execute("CreateObject("Wscript.Shell").Run%%20"powershell.exe%%20-nop%%20-Command%%20Write-Host%%20Hello,%%20MSHTA!;Start-Sleep%%20-Seconds%%205"))</script<"	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	WIN-L5NDHLESDB7\Adminis

Sysmon Event ID: 1 (Process Creation)

Event ID 1 captured the execution of mshta.exe and its associated command-line arguments. This made it possible to identify the use of MSHTA for executing a malicious HTA file.

Attack 3: **T1003.001 (Credential Access) | LSASS Dumping**

The attack attempted to access the LSASS process memory.

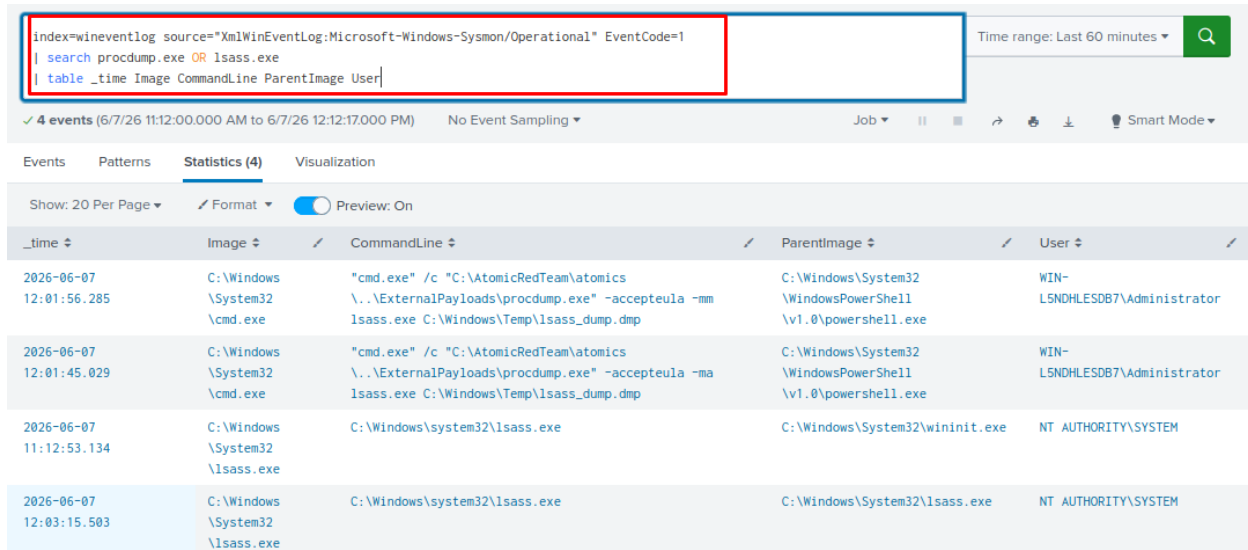


Detection and Query:

```
index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
```

```
| search procdump.exe OR lsass.exe
```

```
| table _time Image CommandLine ParentImage User
```



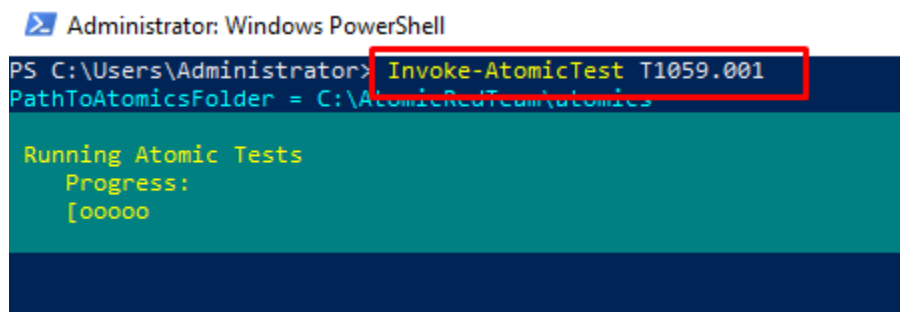
_time	Image	CommandLine	ParentImage	User
2026-06-07 12:01:56.285	C:\Windows\System32\cmd.exe	"cmd.exe" /c "C:\AtomicRedTeam\atomic... \ExternalPayloads\procdump.exe" -accepteula -ma lsass.exe C:\Windows\Temp\lsass_dump.dmp	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	WIN-L5NDHLESDB7\Administrator
2026-06-07 12:01:45.029	C:\Windows\System32\cmd.exe	"cmd.exe" /c "C:\AtomicRedTeam\atomic... \ExternalPayloads\procdump.exe" -accepteula -ma lsass.exe C:\Windows\Temp\lsass_dump.dmp	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	WIN-L5NDHLESDB7\Administrator
2026-06-07 11:12:53.134	C:\Windows\System32\lsass.exe	C:\Windows\system32\lsass.exe	C:\Windows\System32\wininit.exe	NT AUTHORITY\SYSTEM
2026-06-07 12:03:15.503	C:\Windows\System32\lsass.exe	C:\Windows\system32\lsass.exe	C:\Windows\System32\lsass.exe	NT AUTHORITY\SYSTEM

Sysmon Event ID: 1 (Process Creation)

Event ID 1 was the most useful event for detecting the LSASS Dumping attack because it recorded the creation of the process responsible for dumping the LSASS memory. The command line showed the execution of the dump operation and the creation of the file `lsass_dump.dmp`, which provided strong evidence of a credential dumping attempt.

Attack 4: T1059.001 (Execution) | PowerShell Download

PowerShell was used to download and execute a script.

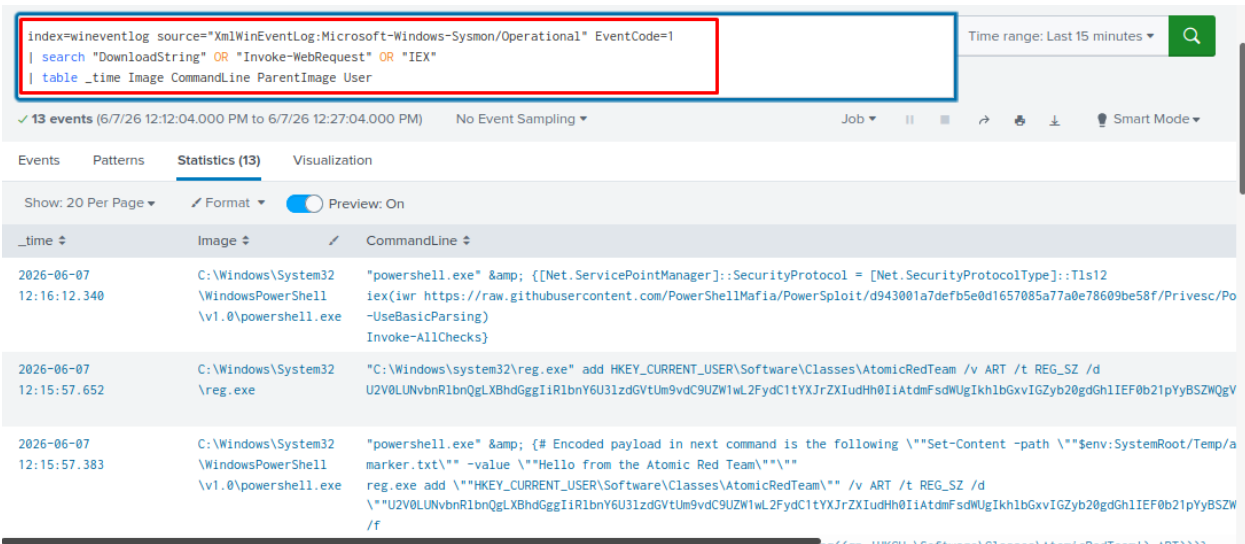


```
Administrator: Windows PowerShell
PS C:\Users\Administrator> Invoke-AtomicTest T1059.001
PathToAtomicFolder = C:\AtomicRedTeam\atomic...

Running Atomic Tests
Progress:
[oooooooo]
```

Detection and Query:

```
index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=1
| search "DownloadString" OR "Invoke-WebRequest" OR "IEX"
| table _time Image CommandLine ParentImage User
```

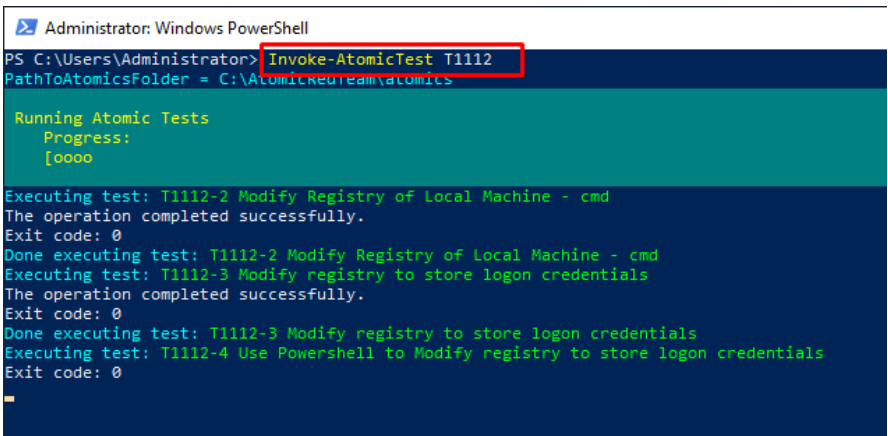


Sysmon Event ID: 1 (Process Creation)

Event ID 1 captured the execution of PowerShell and displayed the command line containing download-related commands such as Invoke-WebRequest, helping identify malicious script execution.

Attack 5: T1112 (Defense Evasion) | Registry Modification

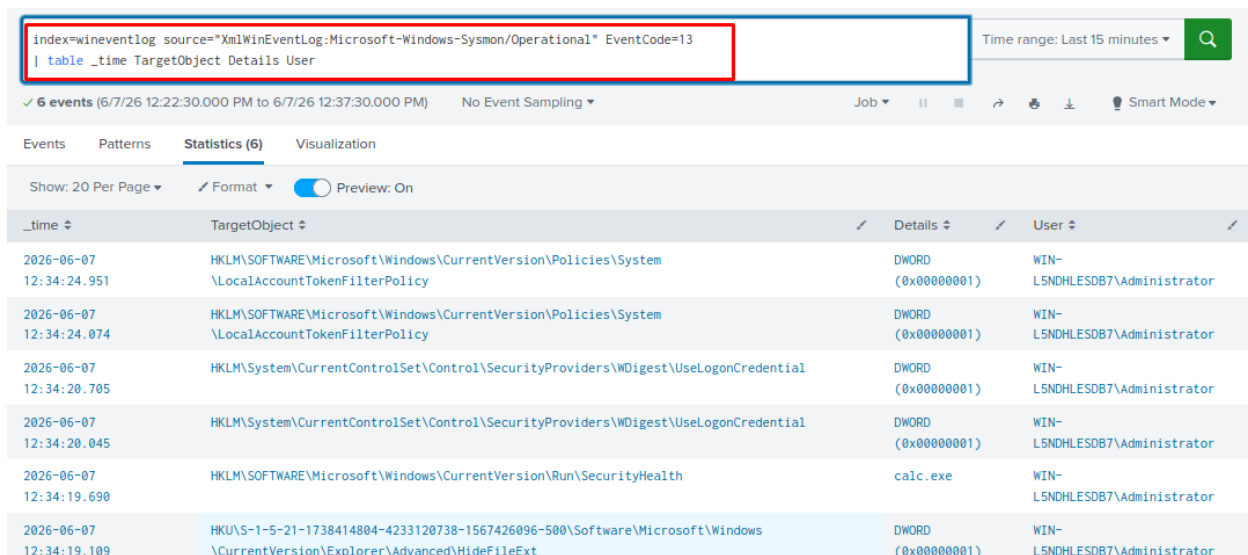
Registry keys were modified to alter system behavior.



Detection and Query:

```
index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"
EventCode=13
```

```
| table _time TargetObject Details User
```



_time	TargetObject	Details	User
2026-06-07 12:34:24.951	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\LocalAccountTokenFilterPolicy	DWORD (0x00000001)	WIN-L5NDHLESDB7\Administrator
2026-06-07 12:34:24.074	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\LocalAccountTokenFilterPolicy	DWORD (0x00000001)	WIN-L5NDHLESDB7\Administrator
2026-06-07 12:34:20.705	HKLM\System\CurrentControlSet\Control\SecurityProviders\WDigest\UseLogonCredential	DWORD (0x00000001)	WIN-L5NDHLESDB7\Administrator
2026-06-07 12:34:20.045	HKLM\System\CurrentControlSet\Control\SecurityProviders\WDigest\UseLogonCredential	DWORD (0x00000001)	WIN-L5NDHLESDB7\Administrator
2026-06-07 12:34:19.690	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SecurityHealth	calc.exe	WIN-L5NDHLESDB7\Administrator
2026-06-07 12:34:19.109	HKU\S-1-5-21-1738414804-4233120738-1567426096-500\Software\Microsoft\Windows\CurrentVersion\Explorer\Advanced\HideFileExt	DWORD (0x00000001)	WIN-L5NDHLESDB7\Administrator

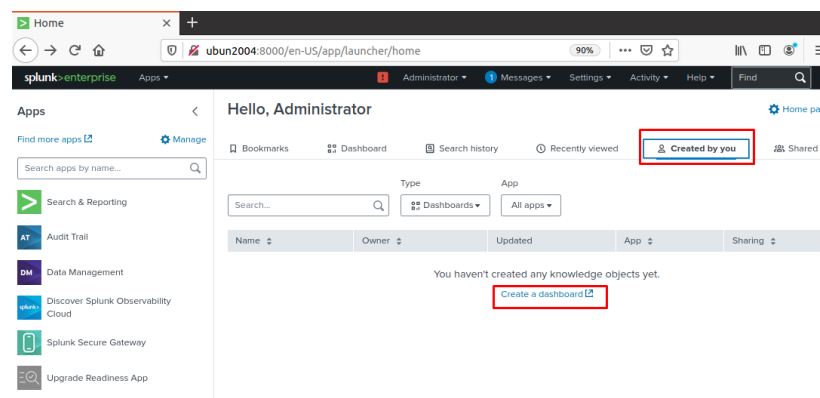
Sysmon Event ID: 13 (Registry Value Set)

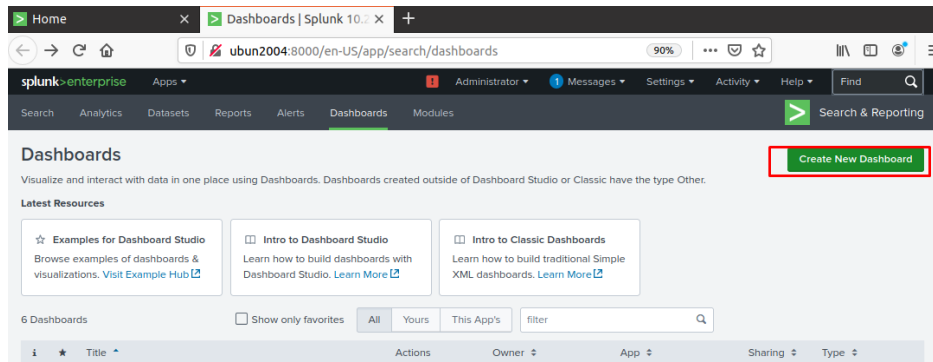
Event ID 13 records registry value modifications. It was the most useful event because it directly captured changes made to registry keys associated with Windows Defender configuration.

5. Splunk Dashboard

A dashboard was created to monitor Sysmon activity and attack indicators.

1. Go to splunk home page and click **“Created by you”**, then click **“create a dashboard”**

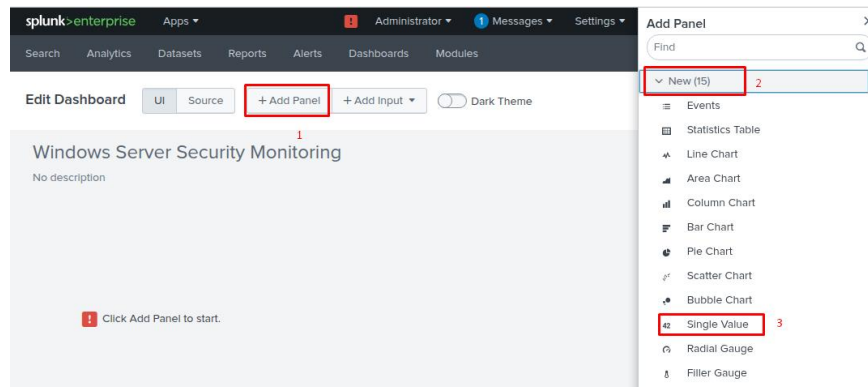




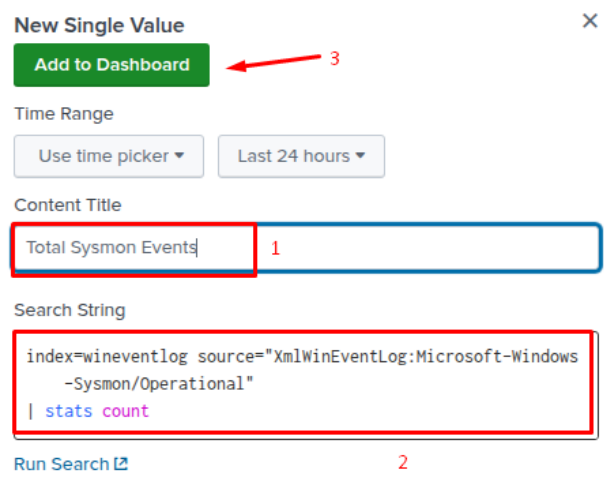
2. Write a title of the Dashboard. Then Scroll down and select a Dashboard type and click create.

A screenshot of the 'Create New Dashboard' dialog box. The 'Dashboard Title' field is filled with 'Windows Server Security Monitoring' and is highlighted with a red rectangle. Below it, the description is 'Optional' and the permissions are set to 'Private'. The 'Dashboard type' section is partially visible, showing a question mark icon. At the bottom, there are 'Cancel' and 'Create' buttons. A red arrow points downwards from the 'Dashboard type' section, indicating the next step.A screenshot of the 'Create New Dashboard' dialog box, showing the 'Dashboard type' section. Two options are presented: 'Classic Dashboards' (The traditional Splunk dashboard builder) and 'Dashboard Studio' (A new builder to create visually-rich, customizable dashboards). Red arrows point to both options, with the word 'or' between them. At the bottom, the 'Create' button is highlighted with a red rectangle.

3. Now click **Add Panel > New > Single Value** and following

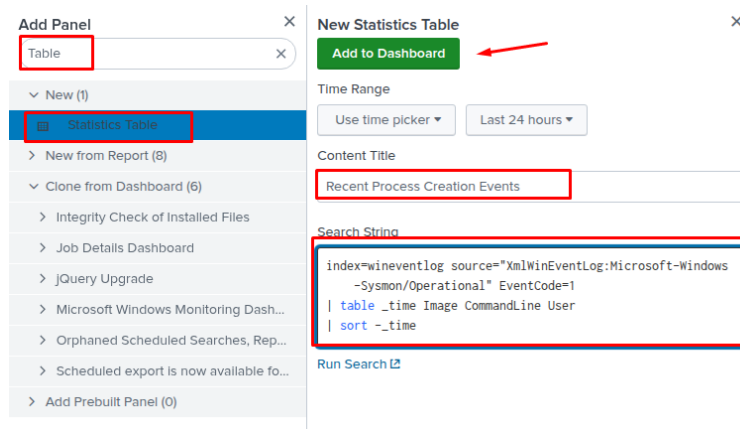


This is show **Total Sysmon Events**



4. Same step again....

This is show **Recent process creation events**



This is show **Sysmon Event Distribution**

The screenshot shows the 'Add Panel' and 'New Pie Chart' configuration windows. In the 'Add Panel' window, 'Pie Chart' is selected under the 'New (15)' category. In the 'New Pie Chart' window, the 'Add to Dashboard' button is highlighted with a red arrow. The 'Content Title' is 'Sysmon Event Distribution'. The 'Search String' is 'index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" | stats count by EventCode'. The 'Time Range' is set to 'Last 24 hours'.

Add Panel

Find

▼ New (15)

- Events
- Statistics Table
- Line Chart
- Area Chart
- Column Chart
- Bar Chart
- Pie Chart**
- Scatter Chart
- Bubble Chart

New Pie Chart

Add to Dashboard

Time Range

Use time picker ▼ Last 24 hours ▼

Content Title

Sysmon Event Distribution

Search String

index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" | stats count by EventCode

Run Search

This is show **Registry Modification Events**

The screenshot shows the 'Add Panel' and 'New Statistics Table' configuration windows. In the 'Add Panel' window, 'Statistics Table' is selected under the 'New (1)' category. In the 'New Statistics Table' window, the 'Add to Dashboard' button is highlighted with a red arrow. The 'Content Title' is 'Registry Modification Events'. The 'Search String' is 'index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=13 | table _time TargetObject Details User'. The 'Time Range' is set to 'Last 24 hours'.

Add Panel

table

▼ New (1)

- Statistics Table**
- > New from Report (8)
- > Clone from Dashboard (6)
- > Integrity Check of Installed Files
- > Job Details Dashboard
- > JQuery Upgrade
- > Microsoft Windows Monitoring Dash...
- > Orphaned Scheduled Searches, Rep...
- > Scheduled export is now available fo...

New Statistics Table

Add to Dashboard

Time Range

Use time picker ▼ Last 24 hours ▼

Content Title

Registry Modification Events

Search String

index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=13 | table _time TargetObject Details User

Run Search

This is show **LSASS Access Attempts**

The screenshot shows the 'Add Panel' and 'New Statistics Table' configuration windows. In the 'Add Panel' window, 'Statistics Table' is selected under the 'New (1)' category. In the 'New Statistics Table' window, the 'Add to Dashboard' button is highlighted with a red arrow. The 'Content Title' is 'LSASS Access Attempts'. The 'Search String' is 'index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 | search procdump.exe OR lsass.exe | table _time Image CommandLine ParentImage User'. The 'Time Range' is set to 'Last 24 hours'.

Add Panel

tab

▼ New (1)

- Statistics Table**
- > New from Report (8)
- > Clone from Dashboard (6)
- > Integrity Check of Installed Files
- > Job Details Dashboard
- > JQuery Upgrade
- > Microsoft Windows Monitoring Dash...
- > Orphaned Scheduled Searches, Rep...
- > Scheduled export is now available fo...
- > Add Prebuilt Panel (0)

New Statistics Table

Add to Dashboard

Time Range

Use time picker ▼ Last 24 hours ▼

Content Title

LSASS Access Attempts

Search String

index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1 | search procdump.exe OR lsass.exe | table _time Image CommandLine ParentImage User

Run Search

This is show **Powershell Activity Events**

Add Panel

tab

New (1)

Statistics Table

New from Report (8)

Bucket Merge Retrieve Conf Settl...

Bulk Data Move Retrieve Conf Se...

Errors in the last 24 hours

Errors in the last hour

License Usage Data Cube

Messages by minute last 3 hours

Orphaned scheduled searches

Splunk errors last 24 hours

Clone from Dashboard (6)

Integrity Check of Installed Files

New Statistics Table

Add to Dashboard

Time Range

Use time picker

Last 24 hours

Content Title

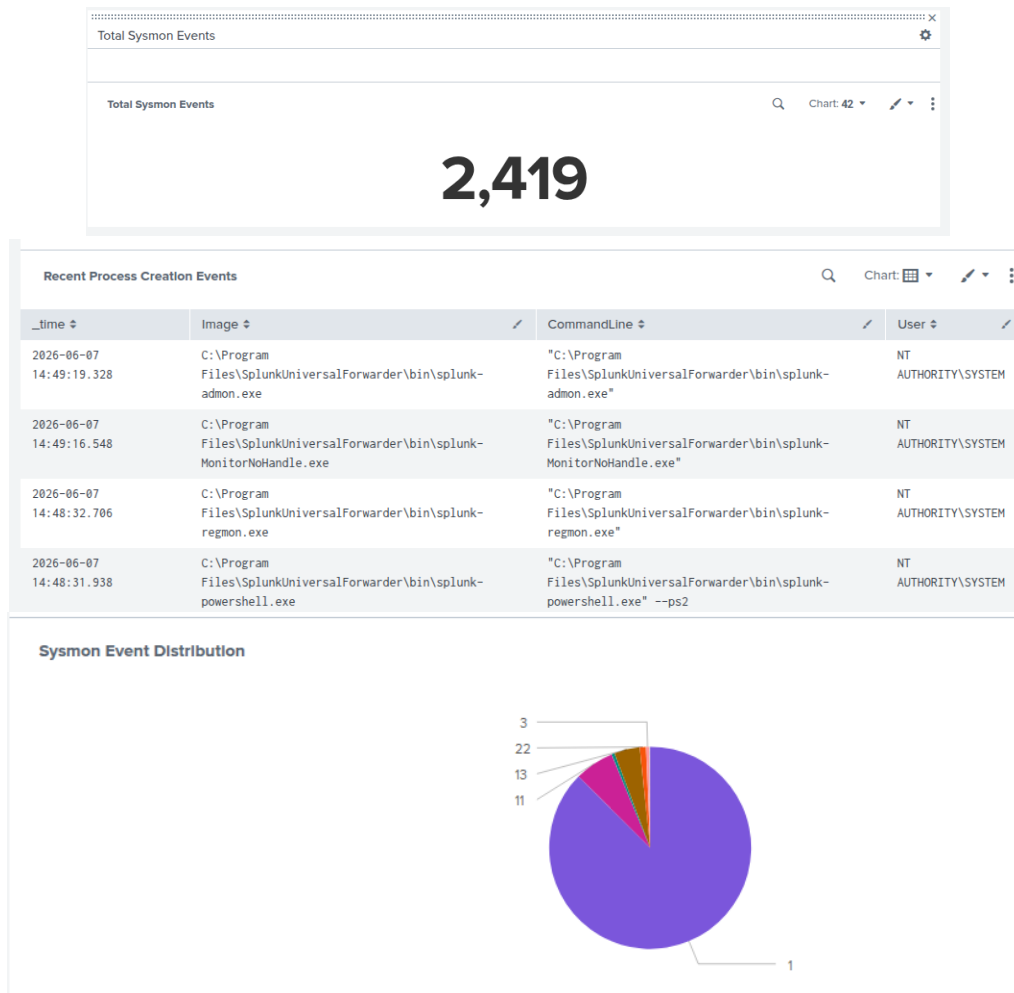
PowerShell Activity

Search String

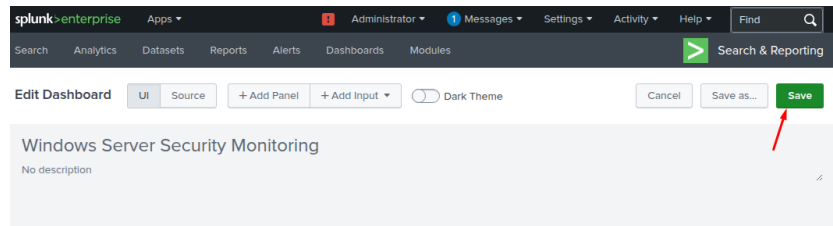
index=wineventlog source="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1
| search "DownloadString" OR "Invoke-WebRequest" OR "IEX"
| table _time Image CommandLine ParentImage User

Run Search

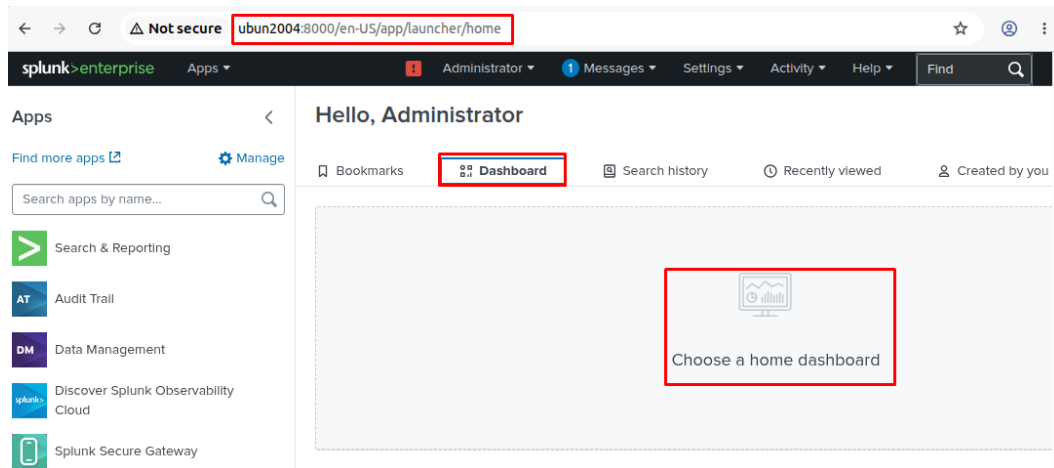
See the Output:



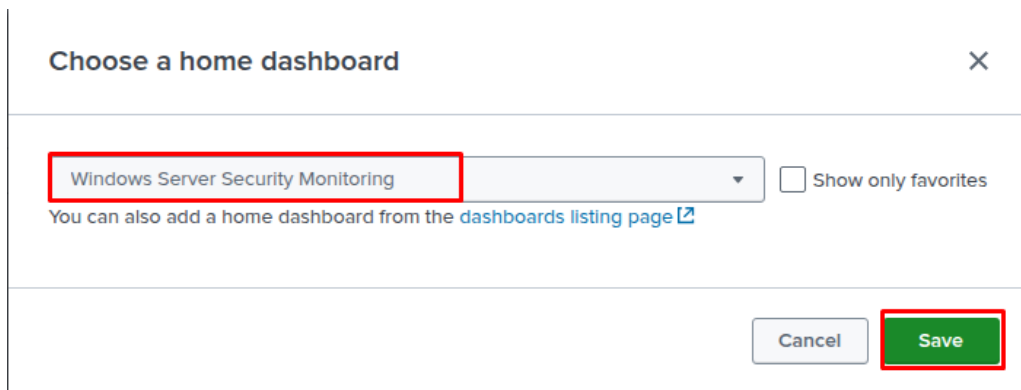
5. Click **Save**



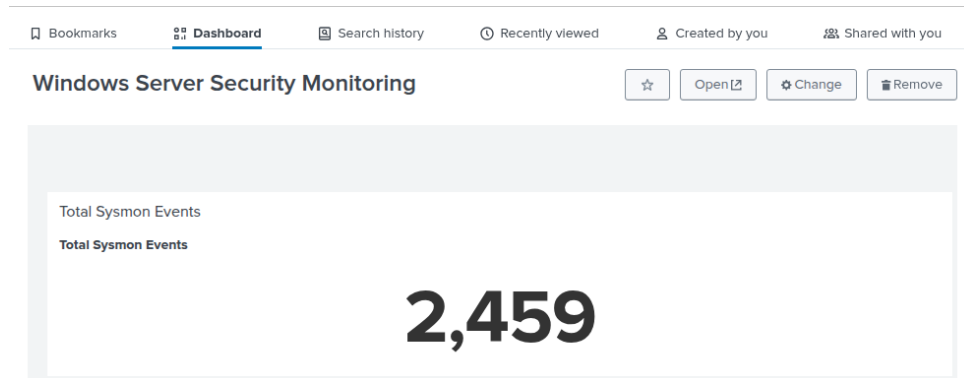
6. Now go to home page, click **Dashboard > Choose a home dashboard**



7. Select created Dashboard and click **Save**



Dashboard Result Show:



7. Challenges and Solutions

- Challenge 1: Sysmon Fields Not Appearing in Splunk

Initially, Sysmon fields such as EventCode, Image, and CommandLine were not displayed correctly in Splunk. This issue was resolved by enabling XML rendering (**renderXml=true**) in the Splunk Universal Forwarder configuration and installing the Splunk Add-on for Microsoft Windows.

- Challenge 2: Event Detection Queries Returning No Results

Some detection queries returned no results because Sysmon events were stored under the **XmlWinEventLog:Microsoft-Windows-Sysmon/Operational** source instead of the standard Windows Event Log source. Updating the search queries to use the correct source resolved the issue.

- Challenge 3: Splunk Dashboard Creation Error

While creating the Splunk dashboard, the error **e.toReversed is not a function** occurred. The issue was caused by using an outdated Firefox browser version. Updating the browser or using a modern browser resolved the problem.

8. Conclusion

This lab demonstrated the deployment of a SOC monitoring environment using Splunk, Sysmon, and Atomic Red Team. Five MITRE ATT&CK techniques were successfully executed and detected using Sysmon event logs. The exercise provided practical experience in attack detection, log analysis and security monitoring.